research note 4.5

1 Mathematica code

viscosity cross section $\int \frac{d\sigma}{d\Omega} d\Omega \sin^2\theta$ from 1709.07882

```
ClearAll["Global`*"];

sKin = 4*(p^2 + m\[Chi]^2);
sNR = 4*m\[Chi]^2;
tKin = -2*p^2*(1 - c);
uKin = -2*p^2*(1 + c);


(*Process 1*)
M1 = -(y*m\[Chi])^2/(sKin - m\[Phi]^2) - (y*m\[Chi])^2/(tKin -
      m\[Phi]^2);
dSigma1 = M1^2/(64*Pi^2*sNR);
integrand1 = 1*2*Pi*(1 - c^2)*dSigma1;

(*Process 2: 1/2 factor for identical particles*)
M2 = -(y*m\[Chi])^2/(tKin - m\[Phi]^2) - (y*m\[Chi])^2/(uKin -
      m\[Phi]^2);
dSigma2 = M2^2/(64*Pi^2*sNR);
integrand2 = 1/2*2*Pi*(1 - c^2)*dSigma2;

sigmaV1Rel =
  Integrate[integrand1, {c, -1, 1},
   Assumptions -> {p > 0, m\[Phi] > 0, m\[Chi] > 0}];

sigmaV2Rel =
  Integrate[integrand2, {c, -1, 1},
   Assumptions -> {p > 0, m\[Phi] > 0, m\[Chi] > 0}];

(*NR Limit and introduce R coefficient*)

nrRules = {y^4 -> (4*Pi*\[Alpha]\[Chi])^2, p -> (m\[Chi]*v/2)};
rRule = {m\[Phi] -> (m\[Chi]*v/R)};

sigmaV1NR =
  Simplify[sigmaV1Rel /. nrRules /. rRule,
   Assumptions -> {R > 0, m\[Chi] > 0, v > 0, \[Alpha]\[Chi] > 0}];
sigmaV2NR =
  Simplify[sigmaV2Rel /. nrRules /. rRule,
   Assumptions -> {R > 0, m\[Chi] > 0, v > 0, \[Alpha]\[Chi] > 0}];

Print["\n====================================="];
Print["\[Sigma]_V (Process 1) NR Mother Formula with R:"];
Print[sigmaV1NR];
```

```
Print["\n\[Sigma]_V (Process 2) NR Mother Formula with R:"];
Print[sigmaV2NR];
Print["=====================================\n"];

(*---Heavy mediator limit,R <<1---*)
sigmaV1HeavyBase =
  Series[sigmaV1NR, {R, 0, 4}] // Normal // FullSimplify;
sigmaV2HeavyBase =
  Series[sigmaV2NR, {R, 0, 4}] // Normal // FullSimplify;

Print["\[Sigma]_V (Process 1, R << 1):"];
Print[Simplify[sigmaV1HeavyBase /. R -> (m\[Chi]*v)/m\[Phi],
   Assumptions -> {m\[Chi] > 0, v > 0, m\[Phi] > 0}]];
Print["\n\[Sigma]_V (Process 2, R << 1):"];
Print[Simplify[sigmaV2HeavyBase /. R -> (m\[Chi]*v)/m\[Phi],
   Assumptions -> {m\[Chi] > 0, v > 0, m\[Phi] > 0}]];
Print["=====================================\n"];


(*---Light mediator limit,R>>1---*)
sigmaV1LightBase =
  Series[sigmaV1NR, {R, Infinity, 0}] // Normal // FullSimplify;
sigmaV2LightBase =
  Series[sigmaV2NR, {R, Infinity, 0}] // Normal // FullSimplify;

Print["\[Sigma]_V (Process 1, R >> 1):"];
Print[Simplify[sigmaV1LightBase /. R -> (m\[Chi]*v)/m\[Phi],
   Assumptions -> {m\[Chi] > 0, v > 0, m\[Phi] > 0}]];
Print["\n\[Sigma]_V (Process 2, R >> 1):"];
Print[Simplify[sigmaV2LightBase /. R -> (m\[Chi]*v)/m\[Phi],
   Assumptions -> {m\[Chi] > 0, v > 0, m\[Phi] > 0}]];
Print["====================================="];
Print["> All calculations finished."];
```

2 Results

NR limit ($v \ll 1$); introduce $R = m_\chi v/m_\phi$

2.1 Mother Formula with R

Process 1

$$\sigma_V^{(1)} = \frac{\pi\alpha_\chi^2}{6m_\chi^2 v^4 \left(v^2 - R^2(v^2+4)\right)^2}\Bigg[- R^4(17v^4 + 120v^2 + 192) + 6R^2v^2(3v^2+8) + 6\left(R^4(v^2+4)^2 + 2R^2(v^2+4)^2 - v^2(3v^2+8)\right)\log(R^2+1)\Bigg] \tag{1}$$

Process 2

$$\sigma_V^{(2)} = \frac{\pi\alpha_\chi^2}{m_\chi^2 v^4}\left[\frac{R^4+2R^2+2}{R^2(R^2+2)}\log(R^2+1) - 1\right] \tag{2}$$

2.2 (Heavy Mediator Limit, $R \ll 1$)

$$\sigma^{(1)}_{V,\text{heavy}} = \frac{2\pi\alpha_\chi^2 m_\chi^2}{3m_\phi^4} \tag{3}$$

$$\sigma^{(2)}_{V,\text{heavy}} = \frac{\pi\alpha_\chi^2 m_\chi^2}{3m_\phi^4} \tag{4}$$

2 factor difference for 1/2 factor for identical particles.

2.3 (Light Mediator Limit, $R \gg 1$)

$$\sigma^{(1)}_{V,\text{light}} = \frac{\pi\alpha_\chi^2}{6m_\chi^2 v^4(v^2+4)^2}\left[12(v^2+4)^2\log\left(\frac{m_\chi v}{m_\phi}\right) - 17v^4 - 120v^2 - 192\right] \tag{5}$$

$$\approx \frac{\pi\alpha_\chi^2}{6m_\chi^2 v^4(v^2+4)^2}\left[12(v^2+4)^2\log\left(\frac{m_\chi v}{m_\phi}\right) - 192\right] \tag{6}$$

$$= \frac{2\pi\alpha_\chi^2}{m_\chi^2 v^4}\log\left(\frac{m_\chi v}{m_\phi}\right) - \frac{32\pi\alpha_\chi^2}{m_\chi^2 v^4(v^2+4)^2} \tag{7}$$

$$= \frac{\pi\alpha_\chi^2}{m_\chi^2 v^4}\left[2\log\left(\frac{m_\chi v}{m_\phi}\right) - \frac{32}{(v^2+4)^2}\right] \tag{8}$$

$$\sigma^{(2)}_{V,\text{light}} = \frac{\pi\alpha_\chi^2}{m_\chi^2 v^4}\left[2\log\left(\frac{m_\chi v}{m_\phi}\right) - 1\right] \tag{9}$$

while in $\sigma^{(1)}$ the v power term should take control, in $\sigma^{(2)}$ the log term should take control.

It is different from the results of σ_T.

But there is log behavior as we expected in 1709.07882's appendix B.

3 Plots

```
ClearAll["Global`*"];

v = 10^-3;
limitSIDM = 4578.4;

(*Heavy,R <<1*)
alpha1Heavy[mChi_, mA_] := Sqrt[(limitSIDM*3*mA^4)/(2*Pi*mChi)];
alpha2Heavy[mChi_, mA_] := Sqrt[(limitSIDM*3*mA^4)/(Pi*mChi)];

(*Light,R>>1*)
denom1Light[mChi_, mA_] :=
  Pi*(12*(v^2 + 4)^2*Log[mChi*v/mA] - 17*v^4 - 120*v^2 - 192);
alpha1Light[mChi_, mA_] :=
  If[denom1Light[mChi, mA] > 0,
   Sqrt[(limitSIDM*6*mChi^3*v^4*(v^2 + 4)^2)/denom1Light[mChi, mA]],
   Indeterminate];

denom2Light[mChi_, mA_] := Pi*(2*Log[mChi*v/mA] - 1);
alpha2Light[mChi_, mA_] :=
```

```
  If[denom2Light[mChi, mA] > 0,
   Sqrt[(limitSIDM*mChi^3*v^4)/denom2Light[mChi, mA]], Indeterminate];

plotMChi[func_, label_, isHeavy_, fixedMA_] :=
  Module[{mChiR1 = fixedMA/v},
   LogLogPlot[
    ConditionalExpression[func[mChi, fixedMA],
     If[isHeavy, mChi <= mChiR1, mChi >= mChiR1]], {mChi, 10^-6, 1},
    PlotRange -> {10^-14, 10}, PlotStyle -> Directive[Thick, Blue],
    Filling -> Top, FillingStyle -> Directive[Opacity[0.15], Black],
    Frame -> True,
    FrameLabel -> {Style[
       "\!\(\*SubscriptBox[\(m\), \(\[Chi]\)]\) [GeV]", 12],
      Style["\!\(\*SubscriptBox[\(\[Alpha]\), \(\[Chi]\)]\)", 12]},
    PlotLabel -> Style[label, 13], GridLines -> {{mChiR1}, None},
    GridLinesStyle -> Directive[Thick, Gray, DotDashed],
    Epilog ->
     If[10^-6 <= mChiR1 <=
       1, {Text[
        Style["R = 1", 12, Bold,
         Darker[Gray]], {mChiR1*If[isHeavy, 0.2, 5], 10^-12}]}, {}],
    ImageSize -> 400]];


plotMA[func_, label_, isHeavy_, fixedMChi_] :=
  Module[{mAR1 = fixedMChi*v},
   LogLogPlot[
    ConditionalExpression[func[fixedMChi, mA],
     If[isHeavy, mA >= mAR1, mA <= mAR1]], {mA, 10^-12, 10^3},
    PlotRange -> {10^-14, 10}, PlotStyle -> Directive[Thick, Red],
    Filling -> Top, FillingStyle -> Directive[Opacity[0.15], Black],
    Frame -> True,
    FrameLabel -> {Style["\!\(\*SubscriptBox[\(M\), \(A\)]\) [GeV]",
       12], Style["\!\(\*SubscriptBox[\(\[Alpha]\), \(\[Chi]\)]\)",
       12]}, PlotLabel -> Style[label, 13],
    GridLines -> {{mAR1}, None},
    GridLinesStyle -> Directive[Thick, Gray, DotDashed],
    Epilog ->
     If[10^-12 <= mAR1 <=
       10^3, {Text[
        Style["R = 1", 12, Bold,
         Darker[Gray]], {mAR1*If[isHeavy, 5, 0.2], 10^-12}]}, {}],
    ImageSize -> 400]];


fixedMAHeavy = 10^-2;
fixedMALight = 10^-7;
fixedMChiAll = 10^-1;

p1 = plotMChi[alpha1Heavy,
   "Proc 1 (Heavy) vs \!\(\*SubscriptBox[\(m\), \(\[Chi]\)]\)", True,
   fixedMAHeavy];
p2 = plotMChi[alpha2Heavy,
   "Proc 2 (Heavy) vs \!\(\*SubscriptBox[\(m\), \(\[Chi]\)]\)", True,
   fixedMAHeavy];

```

```
p3 = plotMA[alpha1Heavy,
   "Proc 1 (Heavy) vs \!\(\*SubscriptBox[\(M\), \(A\)]\)", True,
   fixedMChiAll];
p4 = plotMA[alpha2Heavy,
   "Proc 2 (Heavy) vs \!\(\*SubscriptBox[\(M\), \(A\)]\)", True,
   fixedMChiAll];

p5 = plotMChi[alpha1Light,
   "Proc 1 (Light) vs \!\(\*SubscriptBox[\(m\), \(\[Chi]\)]\)", False,
    fixedMALight];
p6 = plotMChi[alpha2Light,
   "Proc 2 (Light) vs \!\(\*SubscriptBox[\(m\), \(\[Chi]\)]\)", False,
    fixedMALight];

p7 = plotMA[alpha1Light,
   "Proc 1 (Light) vs \!\(\*SubscriptBox[\(M\), \(A\)]\)", False,
   fixedMChiAll];
p8 = plotMA[alpha2Light,
   "Proc 2 (Light) vs \!\(\*SubscriptBox[\(M\), \(A\)]\)", False,
   fixedMChiAll];


GraphicsGrid[{{p1, p2}, {p3, p4}, {p5, p6}, {p7, p8}},
 Spacings -> {10, 10}, ImageSize -> 900]
```

3.1 results

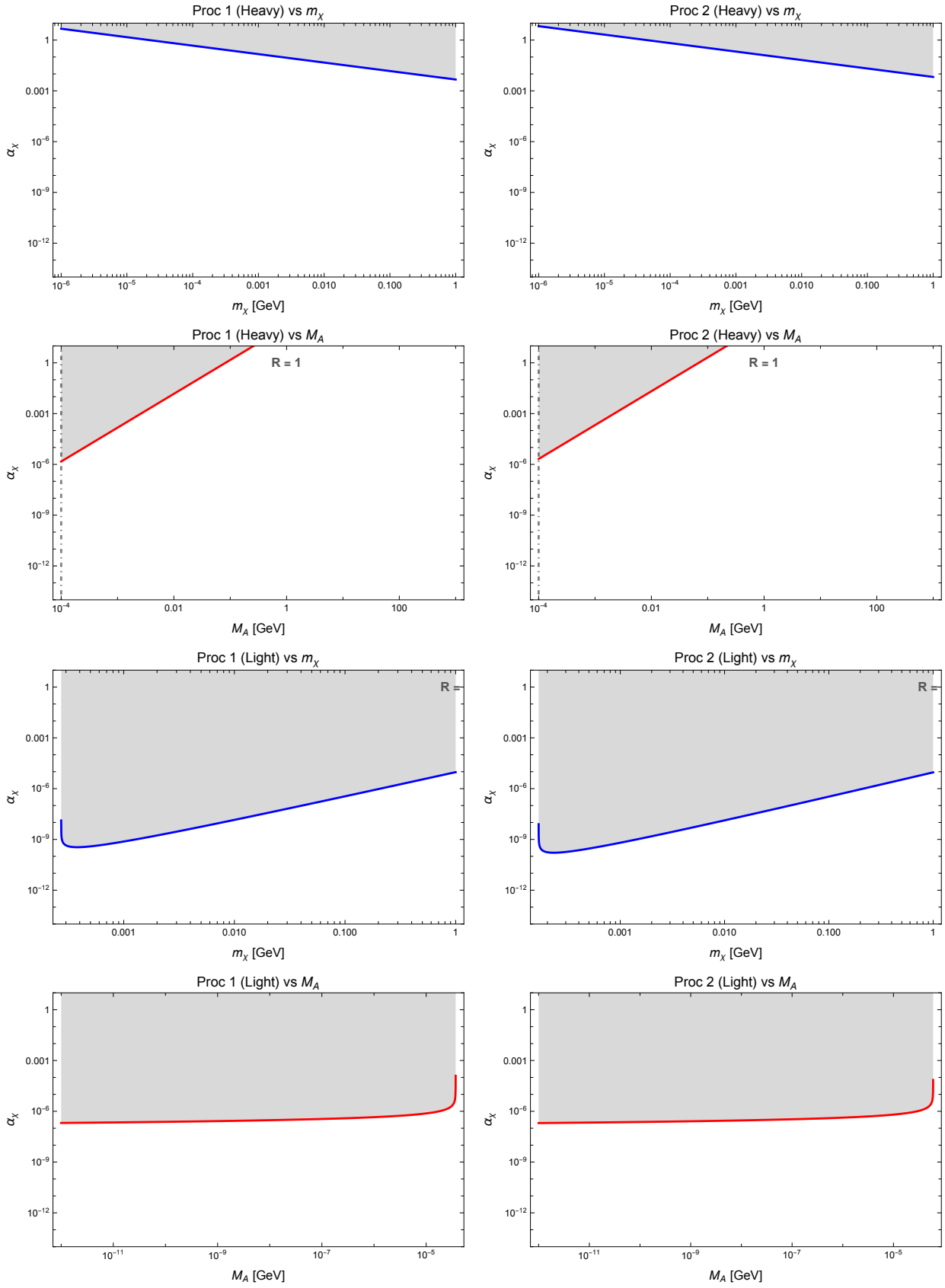


Figure 1:
for heavy mediator I used fixed $M_A = 10^{-2}$(GeV);
for light mediator I used fixed $M_A = 10^{-7}$;
for all $m_\chi = 10^{-1}$;